MILITARY SPECIFICATION

FILTER ELEMENT, AIR, DIVER'S: POLYURETHANE FOAM

1. SCOPE

1.1 This specification covers one type of foam plastic filter element used in a diver's air supply line.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-406 —Plastics, Organic: General Specifications, Test Methods.

CCC-T-191 — Textile Test Methods.

PPP-B-636 — Boxes, Fiberboard.

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MIL-P-116 — Preservation, Methods of.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 601 — Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedure and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

D-1564 — Tentative Specification and Method of Test for Flexible Urethane Foam.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 **Description.** The filter element shall be a skeletal polyurethane foam, Scott Industrial Foam S103-10-30, produced by Scott Paper Company, Foam Division, Chester, Pennsylvania, or equal. The filter element shall be composed entirely of fibers bonded together at two or more junctions to form a lattice and shall have 30 plus or minus 5 pores per linear inch. The foam shall have a density of 1.75 plus or minus 0.1 pounds per cubic foot, a tensile strength of not less than 20 pounds per square inch, an elongation in tension of not less than 220 percent of the original length, and a tear resistance of not less than 3.6 pounds per inch.

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3.2 Preproduction model. The preproduction model is defined as a production model which will be identical to the filter elements which the manufacturer will subsequently produce in fulfillment of the contract. As soon as practicable after award of the contract and prior to submission of any filter elements for final acceptance, the contractor shall furnish a preproduction model for examination and test to determine conformance to this specification. Examination and tests shall be those specified herein. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to supply filter elements conforming to this specification. Any changes or deviations of production filter elements from the preproduction model shall be subject to the approval of the contracting officer.

3.3 Color. The filter element shall be beige.

3.4 Dimensions. The filter element shall have a doughnut shape with an outer diameter of 3¾ inches. A circular hole having a diameter of 1⅜ inches shall be cut from the center of the filter element. The thickness of the filter element shall be 1$\frac{1}{16}$ inches. The tolerance for all dimensions shall be plus or minus $\frac{1}{16}$ inch. The shape shall be obtained by die-cutting, hot-wire cutting, or any other method which will not compress and fuse the foam at its edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection of the filter element.

(b) Production inspection of the filter element.

(c) Inspection of preparation for delivery.

4.3 Preproduction inspection.

4.3.1 *Examination.* The filter element shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 *Tests.* The filter element shall be tested as specified in 4.5.2. Failure to pass any test shall be cause for rejection.

4.4 Production inspection.

4.4.1 *Sampling.*

4.4.1.1 *For examination.* Sampling for examination shall be in accordance with Standard MIL–STD–105.

4.4.1.2 *For tests.* Sampling for test shall be in accordance with Standard MIL–STD–105, inspection level L4.

4.4.2 *Examination.* Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective.

4.4.3 *Tests.* Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2. AQL shall be 10.0 percent defective.

4.5 Inspection procedure.

4.5.1 *Examination.* The filter elements shall be examined for the following defects.

101. Material not as specified.
102. Dimensions not as specified.
103. Color not as specified.

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4.5.2 *Tests.* The filter elements shall be tested as follows.

4.5.2.1 *Pore concentration.* Determine the number of pores per linear inch in accordance with Specification CCC–T–191, method 5050. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.2 *Density.* Determine the density in accordance with Specification L–P–406, method 5012. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.3 *Tensile strength.* Determine the tensile strength in accordance with ASTM Designation D–1564. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.4 *Elongation.* Determine the elongation in accordance with Standard FED. TEST METHOD STD. NO. 601, method 4121. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.5 *Tear resistance.* Determine the tear resistance in accordance with ASTM Designation D–1564, except the jaw speed shall be 20 inches per minute. Nonconformance to 3.1 shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Each filter element shall be preserved in accordance with Specification MIL–P–116, method IC–1.

5.1.2 *Level C.* Each filter element shall be preserved and packaged to assure undamaged delivery from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The filter elements shall be packed in close-fitting boxes conforming to Specification PPP–B–636, class 2, V3c, style optional. The gross weight of each box shall not exceed the weight limitation of the box specification. The boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* The filter elements shall be packed in close-fitting boxes conforming to Specification PPP–B–636, class 2, W5c, style optional, in quantities not to exceed the weight limitations of the box.

5.2.3 *Level C.* The filter elements shall be packed to insure undamaged delivery from the contractor to destination.

5.3 Marking. Packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The filter element is intended for use with Military diving equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

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